

EC2 And EFS Assignment

Create an EFS and connect it to 3 different EC2 instances. Make sure that all instances have different operating system. For instance, Ubuntu, Red Hat Linux and Amazon Linux

Solution:

- 1. Open up the AWS Management console**
- 2. Check for the region [us-east-1(N. Virginia)]**
- 3. Search for EC2 in the search box**
- 4. Click on the instances to go to the EC2 console**
- 5. Click on the launch instances and setup the 1st instance for Ubuntu OS:**
 - a) Name: EC2-Ubuntu**
 - b) AMI: Ubuntu OS [Any free tier for Ubuntu OS]**
 - c) Instance type: t2. micro [free tier eligible]**
 - d) Key-pair: Selected existing key [name: FirstKey]**
 - d) Security Group: Default with inbound rule edited**

Services

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Name and tags

info

Name

EC2-Ubuntu

Add additional tags

Application and OS Images (Amazon Machine Image)

info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

RedHat

SUSE Linux

SUSE

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 22.04 LTS (HVM), SSD Volume Type
ami-0c7217cdde317cfec (64-bit (x86)) / ami-05d47d294c2d19e1 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Summary

Number of instances

info

1

Software Image (AMI)

Canonical, Ubuntu, 22.04 LTS, ...read more
ami-0c7217cdde317cfec

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2

Cancel

Launch instance

Review commands

Services

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Amazon Machine Image (AMI)

Free tier eligible

Ubuntu Server 22.04 LTS (HVM), SSD Volume Type
ami-0c7217cdde317cfec (64-bit (x86)) / ami-05d47d294c2d19e1 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Canonical, Ubuntu, 22.04 LTS, amd64 jammy image build on 2023-12-07

Architecture

AMI ID

64-bit (x86)

ami-0c7217cdde317cfec

Verified provider

Instance type

info

Get advice

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.0716 USD per Hour
On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login)

info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

FirstKey

Create new key pair

Summary

Number of instances

info

1

Software Image (AMI)

Canonical, Ubuntu, 22.04 LTS, ...read more
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Virtual server type (instance type)

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Cancel

Launch instance

Services

Search

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Key pair (login)

info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

FirstKey

Create new key pair

Network settings

info

Edit

Network

info

vpc-0b22f56ff603e7fa1

Subnet

info

No preference (Default subnet in any availability zone)

Auto-assign public IP

info

Enable

Firewall (security groups)

info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Common security groups

info

Select security groups

Summary

Number of instances

info

1

Software Image (AMI)

Canonical, Ubuntu, 22.04 LTS, ...read more
ami-0c7217cdde317cfec

Virtual server type (instance type)

t2.micro

Firewall (security group)

default

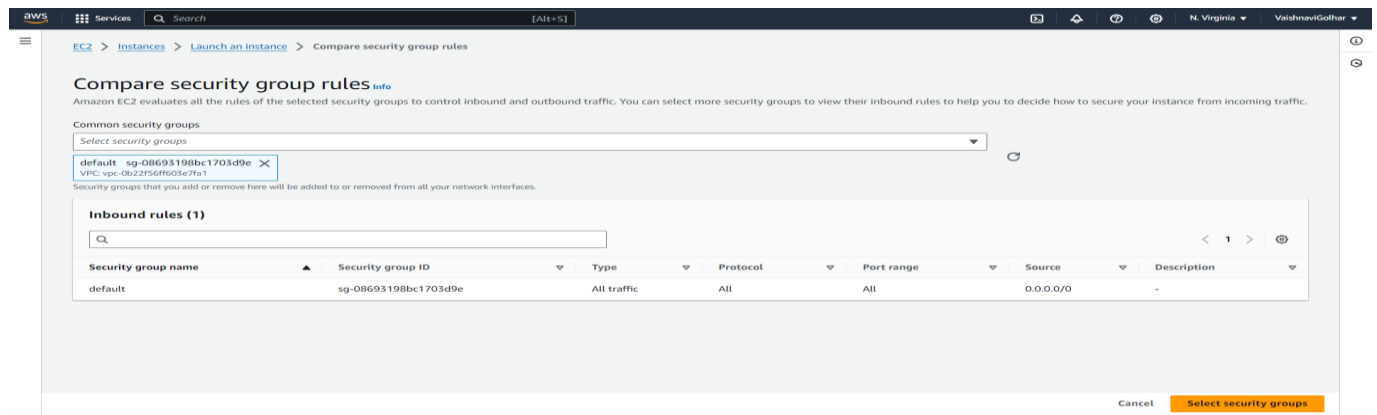
Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2

Cancel

Launch instance



6. Click on launch instance and setup the 2nd instance for Red-Hat-Linux OS:

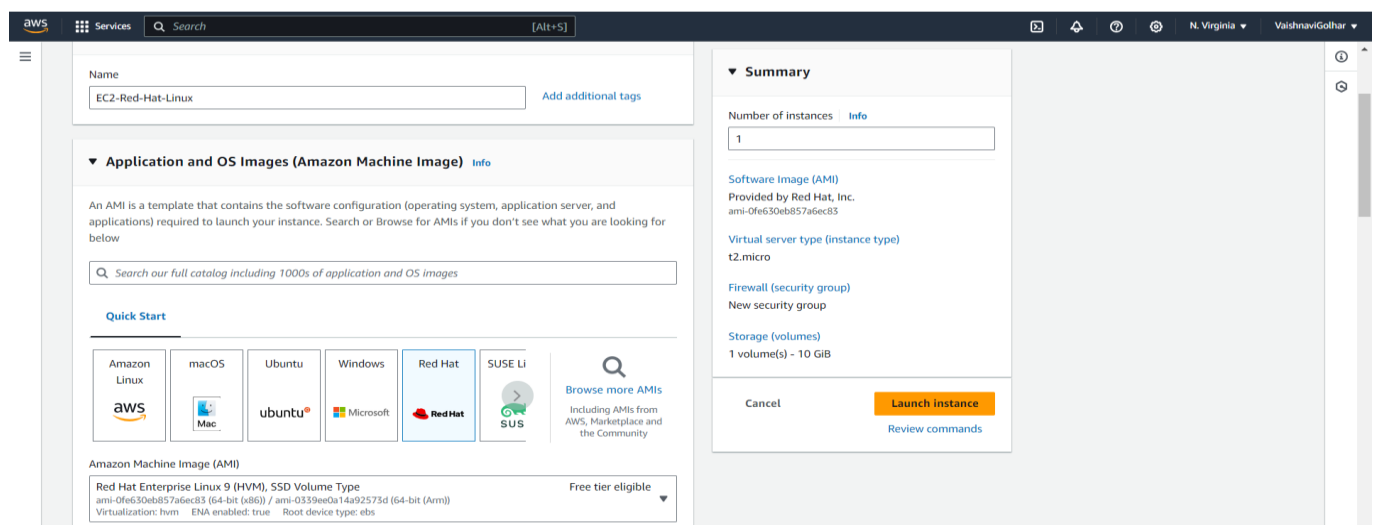
a) Name: EC2-Red-Hat-Linux

b) AMI: Red Hat Linux OS [free tier]

c) Instance type: t2 micro [free tier]

d) Key-pair: Selected existing key [name: FirstKey]

e) Security Group: Default



Description
Provided by Red Hat, Inc.

Architecture
64-bit (x86)

AMI ID
ami-0fe630eb857a6ec83 Verified provider

▼ Instance type [Info](#) | [Get advice](#)

Instance type
t2.micro Free tier eligible
Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.0716 USD per Hour
On-Demand Linux base pricing: 0.0116 USD per Hour
[Compare instance types](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required
FirstKey [Create new key pair](#)

▼ Summary

Number of instances [Info](#)
1

Software Image (AMI)
Provided by Red Hat, Inc.
ami-0fe630eb857a6ec83

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 10 GiB

[Cancel](#) [Launch instance](#) [Review commands](#)

FirstKey [Create new key pair](#)

▼ Network settings [Info](#) [Edit](#)

Network [Info](#)
vpc-0b22f56ff603e7fa1

Subnet [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)
Enable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)
[Select security groups](#)
default sg-08693198bc1703d9e [X](#)
VPC: vpc-0b22f56ff603e7fa1
[Compare security group rules](#)
Security groups that you add or remove here will be added to or removed from all your network interfaces.

▼ Configure storage [Info](#) [Advanced](#)

▼ Summary

Number of instances [Info](#)
1

Software Image (AMI)
Provided by Red Hat, Inc.
ami-0fe630eb857a6ec83

Virtual server type (instance type)
t2.micro

Firewall (security group)
default

Storage (volumes)
1 volume(s) - 10 GiB

[Cancel](#) [Launch instance](#) [Review commands](#)

7. Click on Launch Instance and setup the 3rd instance for Amazon Linux Machine OS:

a) Name: EC2-Amazon Linux Machine

b) AMI: Amazon Linux Machine OS [free tier]

c) Instance type: t2 micro [free tier]

d) Key-pair: Selected existing key [name: FirstKey]

e) Security Group: Default

Name and tags [Info](#)

Name
EC2-Amazon Linux Machine [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Enterprise Server

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 AMI
ami-0440d3b780d96b29d (64-bit (x86), uefi-preferred) / ami-0f93c02efd1974b8b (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs **Free tier eligible**

Summary

Number of instances [Info](#)
1

[Software Image \(AMI\)](#)
Amazon Linux 2023 AMI 2023.3.2...[read more](#)
ami-0440d3b780d96b29d

[Virtual server type \(instance type\)](#)
t2.micro

[Firewall \(security group\)](#)
New security group

[Storage \(volumes\)](#)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Review commands](#)

Description

Amazon Linux 2023 AMI 2023.3.20240219.0 x86_64 HVM kernel-6.1

Architecture: 64-bit (x86) Boot mode: uefi-preferred AMI ID: ami-0440d3b780d96b29d **Verified provider**

Instance type [Info](#) [Get advice](#)

Instance type: t2.micro **Free tier eligible**

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.0716 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required
FirstKey [Create new key pair](#)

Summary

Number of instances [Info](#)
1

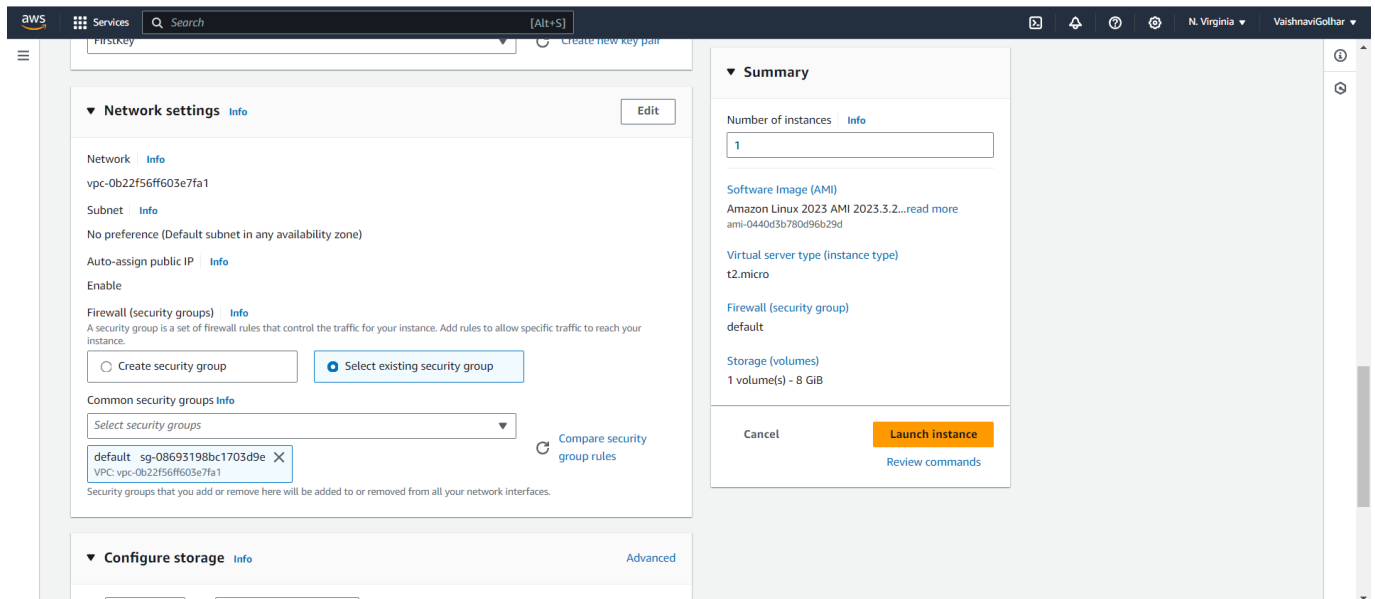
[Software Image \(AMI\)](#)
Amazon Linux 2023 AMI 2023.3.2...[read more](#)
ami-0440d3b780d96b29d

[Virtual server type \(instance type\)](#)
t2.micro

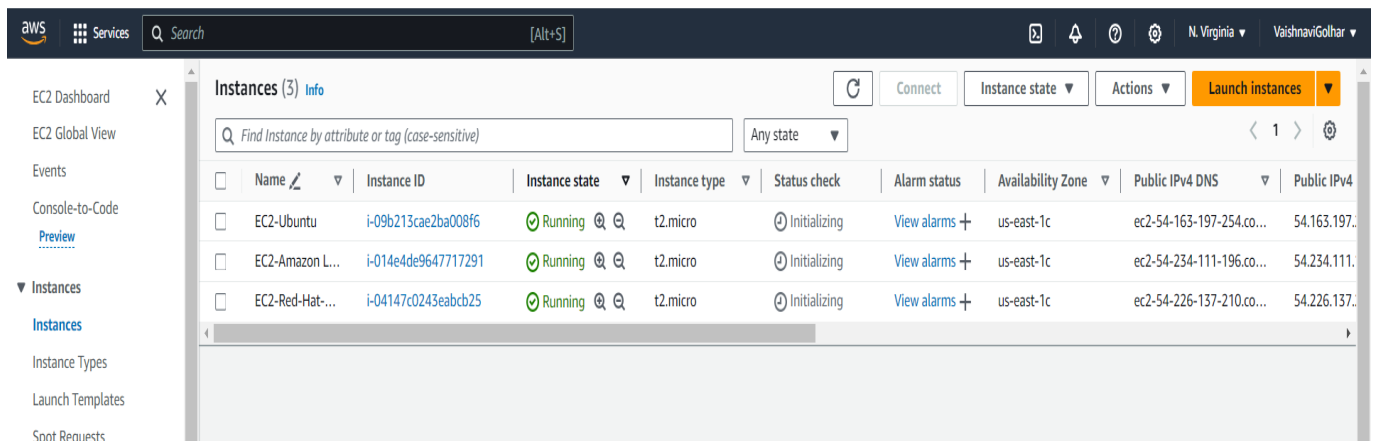
[Firewall \(security group\)](#)
New security group

[Storage \(volumes\)](#)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Review commands](#)



8. Check all the instances in running conditions

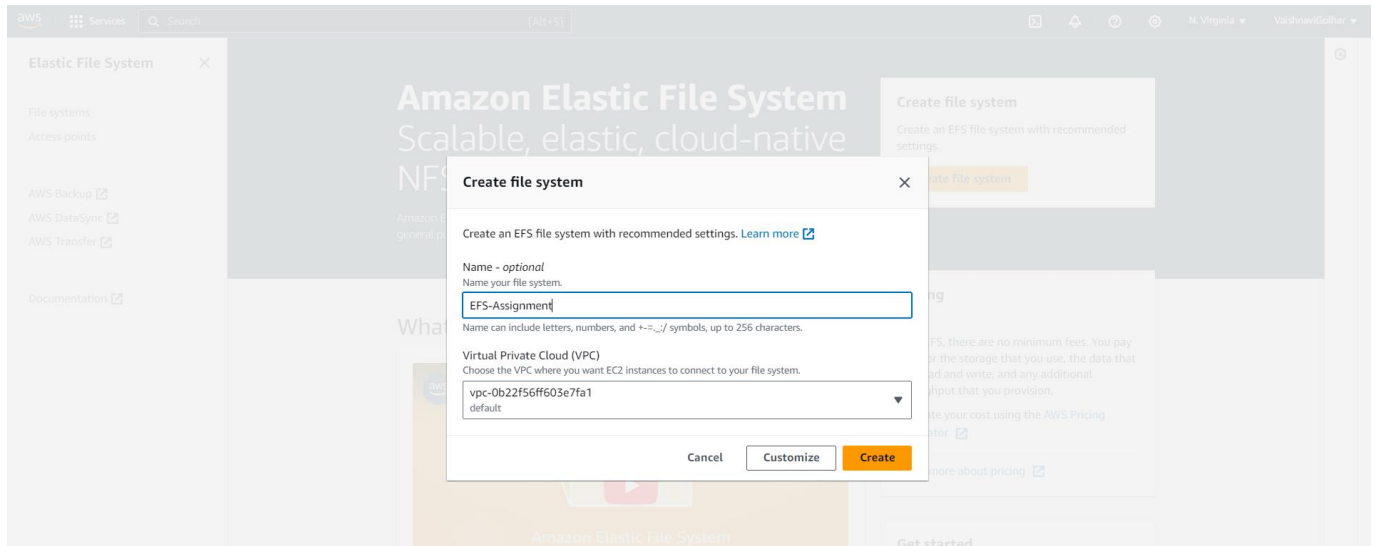


9. Now create a EFS Volume (Elastic File System) in AWS.

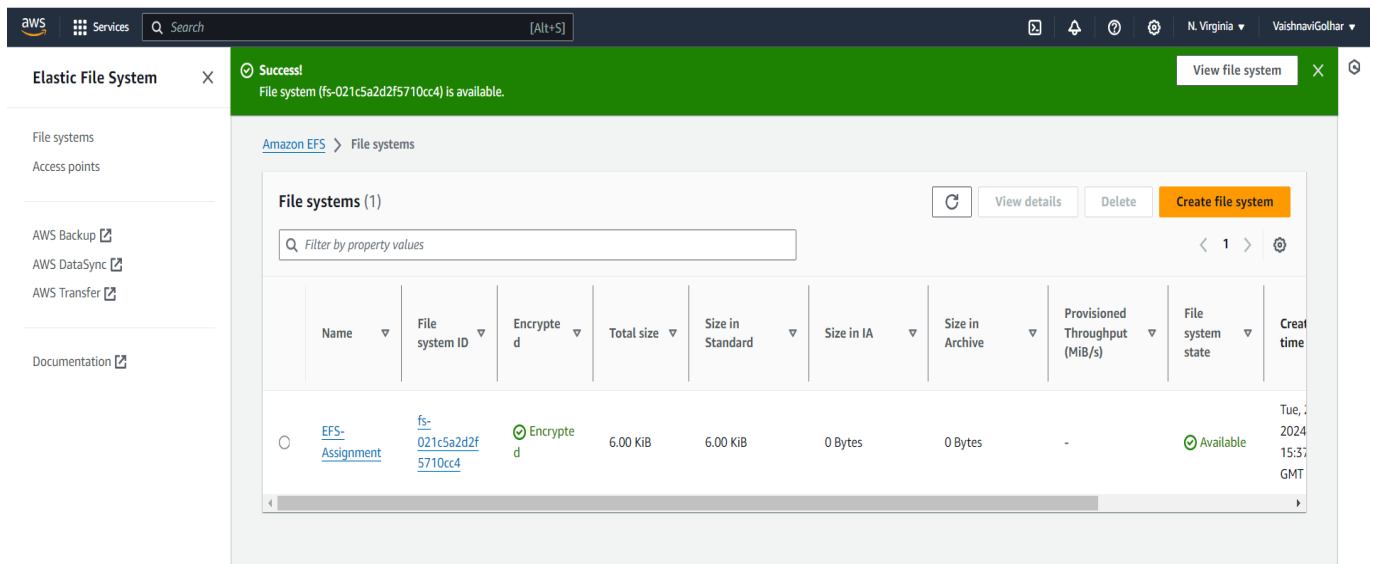
a) Search for EFS in search bar

b) Click on the create file system with the name

EFS-Assignment with default VPC.



3. EFS is now created and is available now.



11. Now will attach the EFS to all 3 instances one by one.

12. EC2-Ubuntu

a) Connect EC2-Ubuntu instance

b) Update sudo apt-get update

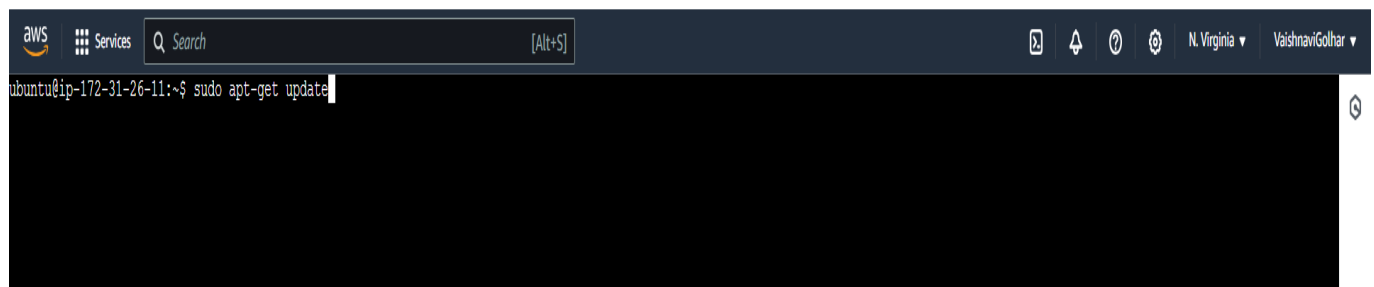
c) Install NFS utils by the command line

`sudo apt install nfs-common`

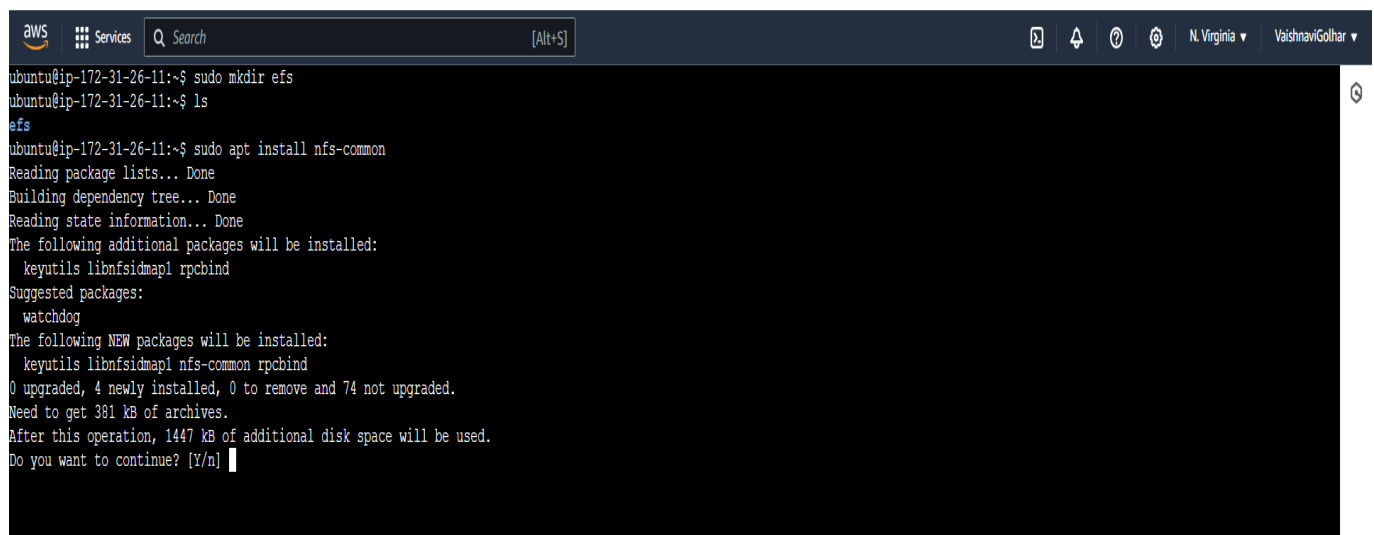
d) `sudo mkdir efs, ls`

e) Mount command from EFS attach

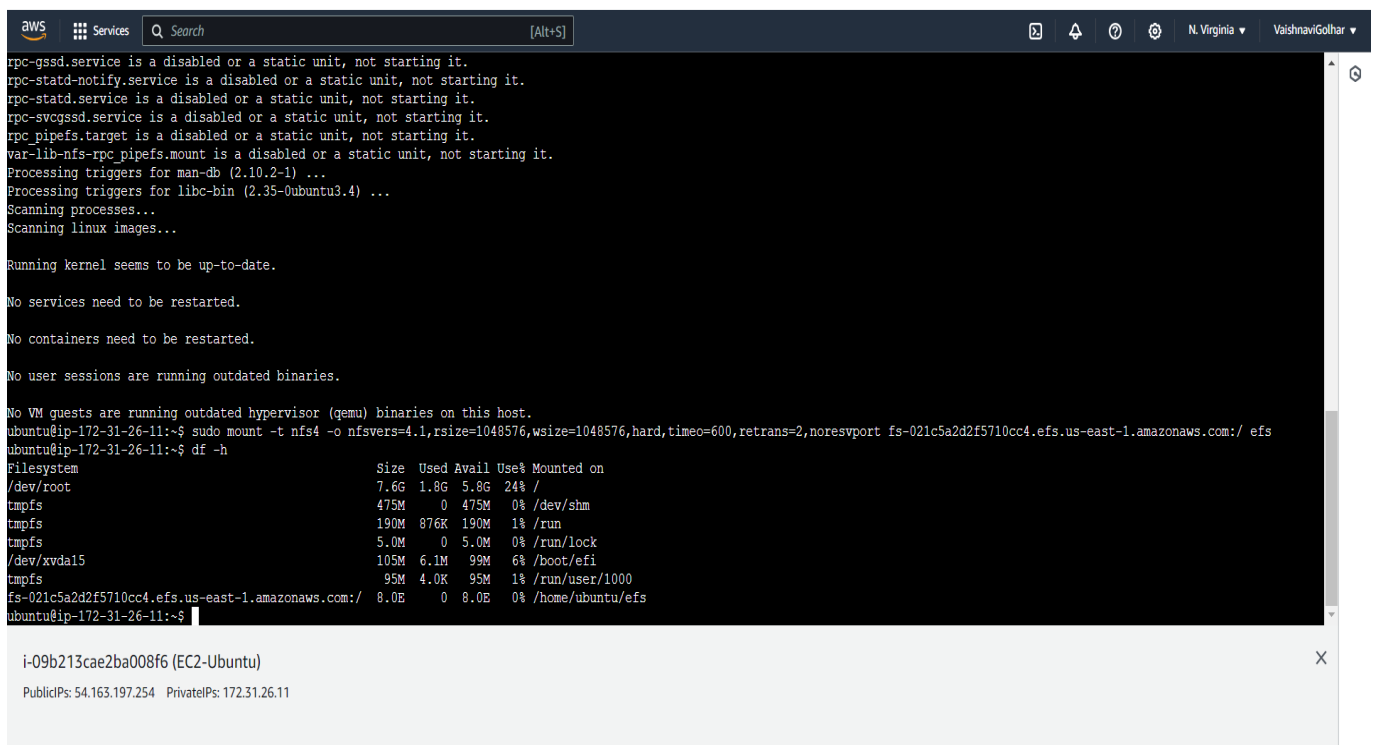
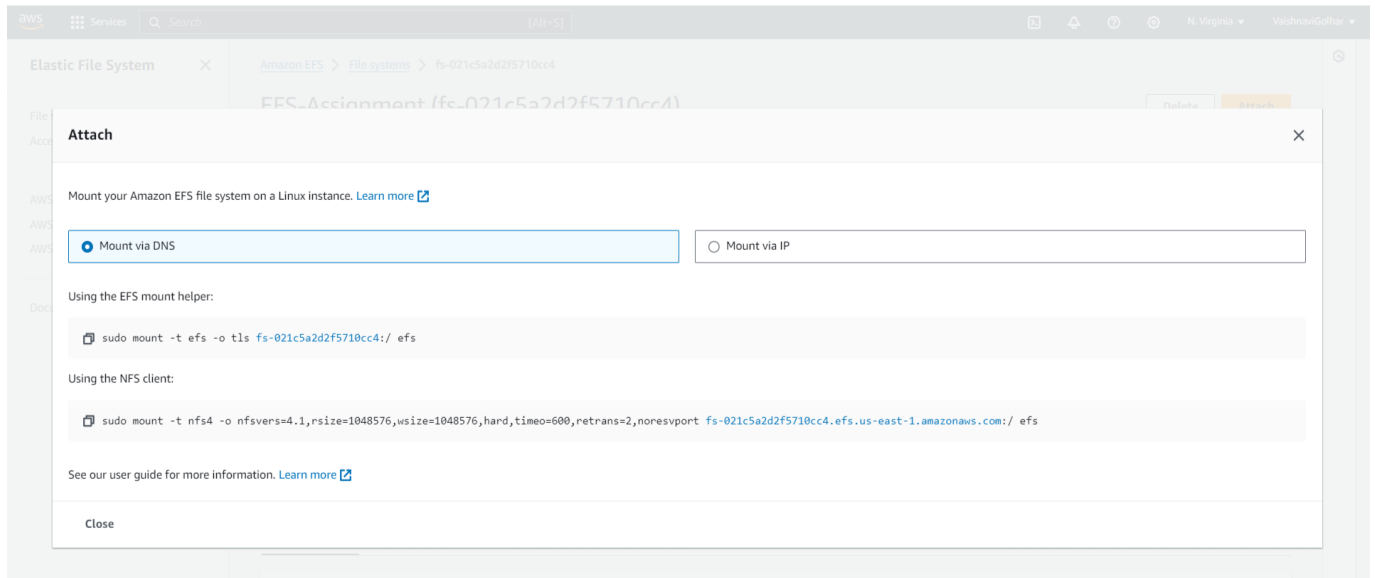
f) `df -h`



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ubuntu@ip-172-31-26-11:~$ sudo apt-get update
```



```
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ubuntu@ip-172-31-26-11:~$ sudo mkdir efs
ubuntu@ip-172-31-26-11:~$ ls
efs
ubuntu@ip-172-31-26-11:~$ sudo apt install nfs-common
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  keyutils libnfsidmap1 rpcbind
Suggested packages:
  watchdog
The following NEW packages will be installed:
  keyutils libnfsidmap1 nfs-common rpcbind
0 upgraded, 4 newly installed, 0 to remove and 74 not upgraded.
Need to get 381 kB of archives.
After this operation, 1447 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

13. EC2-Amazon Linux Machine

a) Connect EC2-Amazon Linux Machine instance

b) Update sudo yum update

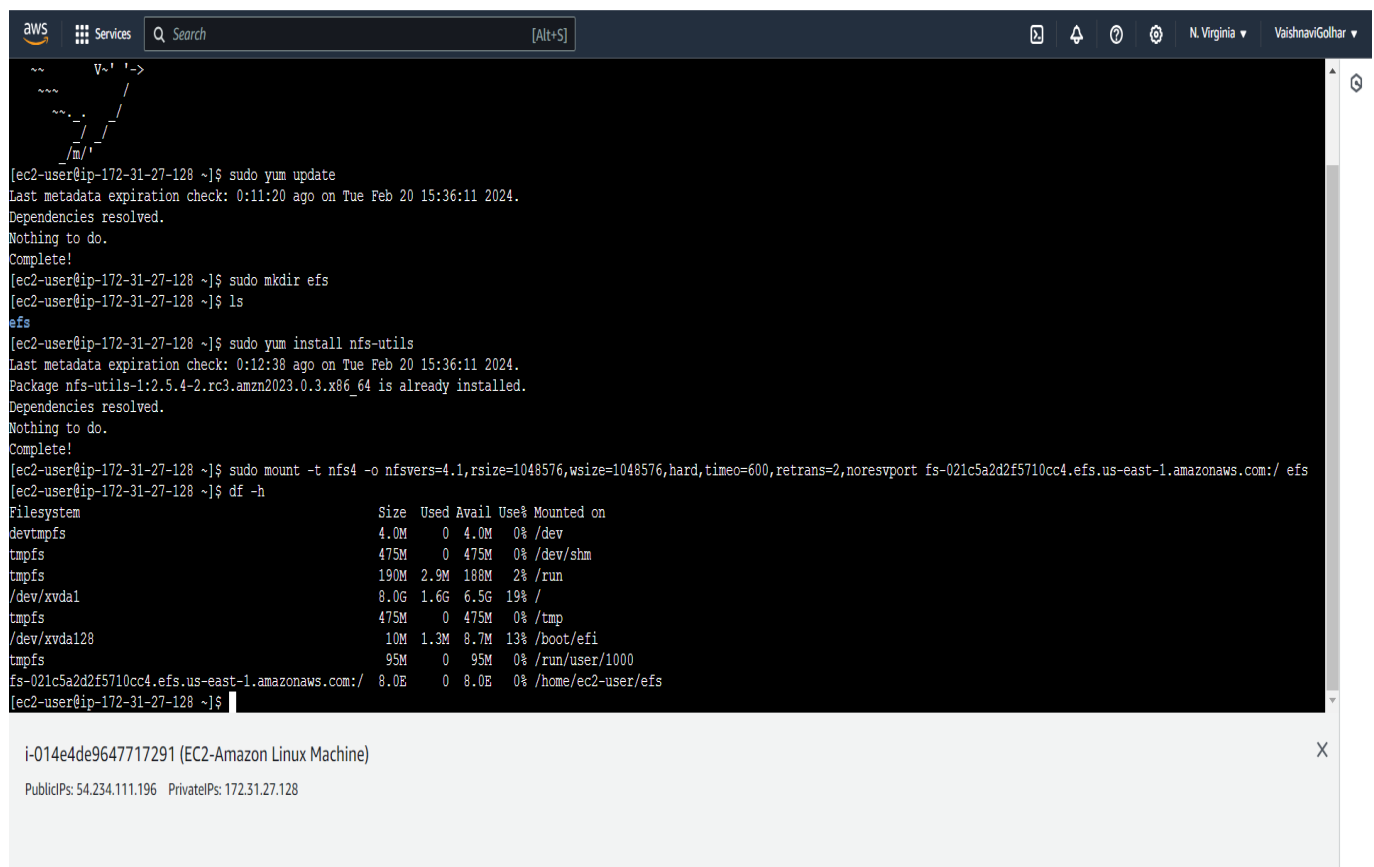
c) Install NFS utils by the command line

sudo yum install nfs-utils

d) sudo mkdir efs, ls

e) Mount command from EFS attach

f) df -h



```
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[ec2-user@ip-172-31-27-128 ~]$ sudo yum update
Last metadata expiration check: 0:11:20 ago on Tue Feb 20 15:36:11 2024.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-27-128 ~]$ sudo mkdir efs
[ec2-user@ip-172-31-27-128 ~]$ ls
efs
[ec2-user@ip-172-31-27-128 ~]$ sudo yum install nfs-utils
Last metadata expiration check: 0:12:38 ago on Tue Feb 20 15:36:11 2024.
Package nfs-utils-1:2.5.4-2.rc3.amzn2023.0.3.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-27-128 ~]$ sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsz=1048576,hard,timeo=600,retrans=2,noresvport fs-021c5a2d2f5710cc4.efs.us-east-1.amazonaws.com:/ efs
[ec2-user@ip-172-31-27-128 ~]$ df -h
Filesystem                                Size  Used Avail Use% Mounted on
devtmpfs                                  4.0M   0   4.0M   0% /dev
tmpfs                                     475M   0   475M   0% /dev/shm
tmpfs                                     190M  2.9M  188M   2% /run
/dev/xvda1                                8.0G  1.6G   6.5G  19% /
tmpfs                                     475M   0   475M   0% /tmp
/dev/xvda128                              10M   1.3M   8.7M  13% /boot/efi
tmpfs                                     95M   0    95M   0% /run/user/1000
fs-021c5a2d2f5710cc4.efs.us-east-1.ama  8.0E   0   8.0E   0% /home/ec2-user/efs
[ec2-user@ip-172-31-27-128 ~]$
```

i-014e4de9647717291 (EC2-Amazon Linux Machine)

PublicIPs: 54.234.111.196 PrivateIPs: 172.31.27.128

13. EC2-Red-Hat-Linux

a) Connect EC2-Red-Hat-Linux instance via Putty

b) Update sudo yum update

c) Install NFS utils by the command line

sudo yum install nfs-utils

d) sudo mkdir efs, ls

e) Mount command from EFS attach

f) df -h

The screenshot shows the AWS Management Console interface. On the left, there is a navigation menu with options like Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Images, AMIs, and AMI Catalog. The main content area displays the details for an EC2 instance named 'i-0798ec1d5c9e8c05a (EC2-Red-Hat-Linux)'. The instance is in a 'Running' state. Key details include: Instance ID: i-0798ec1d5c9e8c05a (EC2-Red-Hat-Linux), Public IPv4 address: 3.80.136.238, Private IPv4 addresses: 172.31.24.127, and Public IPv4 DNS: ec2-3-80-136-238.compute-1.amazonaws.com.

The screenshot shows the AWS Management Console 'Connect to instance' page for the same EC2 instance. It offers four connection methods: EC2 Instance Connect (selected), Session Manager, SSH client, and EC2 serial console. The 'EC2 Instance Connect' method is highlighted, showing the public IP address 3.80.136.238. Overlaid on the right is a 'PuTTY Configuration' window. The 'Basic options for your PuTTY session' tab is active, showing 'Host Name (or IP address)' as 3.80.136.238 and 'Port' as 22. The 'Connection type' is set to 'SSH'. The 'Close window on exit' option is set to 'Only on clean exit'. The 'Open' button is highlighted.

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EC2 > Instances > i-0798ec1d5c9e8c05a > Connect to instance

Connect to instance [Info](#)

Connect to your instance i-0798ec1d5c9e8c05a (EC2-Red-Hat-Linux) using any of these options

EC2 Instance Connect

Session Manager

SSH client

EC2 serial console

Instance ID

i-0798ec1d5c9e8c05a (EC2-Red-Hat-Linux)

Connection Type

☒ Connect using EC2 Instance Connect

☐ Connect using EC2 Instance Connect Endpoint

Connect using the EC2 Instance Connect browser-based client, with a public IPv4 address.

Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

Public IP address

3.80.136.238

PuTTY Configuration

Category: Logging, Terminal, Keyboard, Bell, Features, Window, Appearance, Behaviour, Translation, Selection, Colours, Connection, Data, Proxy, SSH, Kex, Host keys, Cipher, Auth, Credentials, TTY, X11, Tunnels

Credentials to authenticate with

Public-key authentication

Private key file for authentication: F:\AWS\key pair\ABC.ppk

Certificate to use with the private key (optional):

Plugin to provide authentication responses

Plugin command to run:

About, Help, Open, Cancel

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EC2 > Instances > i-0798ec1d5c9e8c05a > Connect to instance

Connect to instance [Info](#)

Connect to your instance i-0798ec1d5c9e8c05a (EC2-Red-Hat-Linux) using any of these options

EC2 Instance Connect

Session Manager

SSH client

EC2 serial console

Instance ID

i-0798ec1d5c9e8c05a (EC2-Red-Hat-Linux)

Connection Type

☒ Connect using EC2 Instance Connect

☐ Connect using EC2 Instance Connect Endpoint

ec2-user@ip-172-31-24-127:~

login as: ec2-user

Authenticating with public key "imported-openssh-key"

Register this system with Red Hat Insights: insights-client --register

Create an account or view all your systems at https://red.ht/insights-dashboard

[ec2-user@ip-172-31-24-127 ~]\$ sudo yum update

Updating Subscription Management repositories.

Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to register.

Red Hat Enterprise Linux 9 for x86_64 - AppStream 45 MB/s | 29 MB 00:00

```
ec2-user@ip-172-31-24-127:~
Cleanup      : rh-amazon-rhui-client-4.0.13-1.el9.noarch 20/28
Cleanup      : tzdata-2023d-1.el9.noarch 21/28
Cleanup      : nss-sysinit-3.90.0-6.el9_3.x86_64 22/28
Cleanup      : nss-softoken-freebl-3.90.0-4.el9_3.x86_64 23/28
Cleanup      : nss-util-3.90.0-4.el9_3.x86_64 24/28
Cleanup      : nspr-4.35.0-6.el9_3.x86_64 25/28
Cleanup      : openssl-lib-1.3.0.7-24.el9.x86_64 26/28
Cleanup      : sudo-1.9.5p2-9.el9.x86_64 27/28
Cleanup      : gnutls-3.7.6-23.el9.x86_64 28/28
Running scriptlet: nss-3.90.0-6.el9_3.x86_64 28/28
Running scriptlet: selinux-policy-targeted-38.1.23-1.el9_3.2.noarch 28/28
Running scriptlet: gnutls-3.7.6-23.el9.x86_64 28/28
Verifying     : nspr-4.35.0-6.el9_3.x86_64 1/28
Verifying     : nspr-4.35.0-6.el9_3.x86_64 2/28
Verifying     : nss-3.90.0-6.el9_3.x86_64 3/28
Verifying     : nss-3.90.0-4.el9_3.x86_64 4/28
Verifying     : nss-softoken-3.90.0-6.el9_3.x86_64 5/28
Verifying     : nss-softoken-3.90.0-4.el9_3.x86_64 6/28
Verifying     : nss-softoken-freebl-3.90.0-6.el9_3.x86_64 7/28
Verifying     : nss-softoken-freebl-3.90.0-4.el9_3.x86_64 8/28
Verifying     : nss-sysinit-3.90.0-6.el9_3.x86_64 9/28
Verifying     : nss-sysinit-3.90.0-4.el9_3.x86_64 10/28
Verifying     : nss-util-3.90.0-6.el9_3.x86_64 11/28
Verifying     : nss-util-3.90.0-4.el9_3.x86_64 12/28
Verifying     : openssl-1.3.0.7-25.el9_3.x86_64 13/28
Verifying     : openssl-1.3.0.7-24.el9.x86_64 14/28
Verifying     : openssl-lib-1.3.0.7-25.el9_3.x86_64 15/28
Verifying     : openssl-lib-1.3.0.7-24.el9.x86_64 16/28
Verifying     : gnutls-3.7.6-23.el9_3.x86_64 17/28
Verifying     : gnutls-3.7.6-23.el9.x86_64 18/28
Verifying     : tzdata-2024a-1.el9.noarch 19/28
Verifying     : tzdata-2023d-1.el9.noarch 20/28
Verifying     : selinux-policy-38.1.23-1.el9_3.2.noarch 21/28
Verifying     : selinux-policy-38.1.23-1.el9_3.1.noarch 22/28
Verifying     : selinux-policy-targeted-38.1.23-1.el9_3.2.noarch 23/28
Verifying     : selinux-policy-targeted-38.1.23-1.el9_3.1.noarch 24/28
Verifying     : sudo-1.9.5p2-10.el9_3.x86_64 25/28
Verifying     : sudo-1.9.5p2-9.el9.x86_64 26/28
Verifying     : rh-amazon-rhui-client-4.0.14-1.el9.noarch 27/28
Verifying     : rh-amazon-rhui-client-4.0.13-1.el9.noarch 28/28
Installed products updated.

Upgraded:
gnutls-3.7.6-23.el9_3.x86_64 nspr-4.35.0-6.el9_3.x86_64 nss-3.90.0-6.el9_3.x86_64 nss-softoken-3.90.0-6.el9_3.x86_64
nss-softoken-freebl-3.90.0-6.el9_3.x86_64 nss-sysinit-3.90.0-6.el9_3.x86_64 nss-util-3.90.0-6.el9_3.x86_64 openssl-1.3.0.7-25.el9_3.x86_64
openssl-lib-1.3.0.7-25.el9_3.x86_64 rh-amazon-rhui-client-4.0.14-1.el9.noarch selinux-policy-38.1.23-1.el9_3.2.noarch selinux-policy-targeted-38.1.23-1.el9_3.2.noarch
sudo-1.9.5p2-10.el9_3.x86_64 tzdata-2024a-1.el9.noarch

Complete!
[ec2-user@ip-172-31-24-127 ~]$
```

```
ec2-user@ip-172-31-24-127:~
Verifying     : nss-sysinit-3.90.0-6.el9_3.x86_64 9/28
Verifying     : nss-sysinit-3.90.0-4.el9_3.x86_64 10/28
Verifying     : nss-util-3.90.0-6.el9_3.x86_64 11/28
Verifying     : nss-util-3.90.0-4.el9_3.x86_64 12/28
Verifying     : openssl-1.3.0.7-25.el9_3.x86_64 13/28
Verifying     : openssl-1.3.0.7-24.el9.x86_64 14/28
Verifying     : openssl-lib-1.3.0.7-25.el9_3.x86_64 15/28
Verifying     : openssl-lib-1.3.0.7-24.el9.x86_64 16/28
Verifying     : gnutls-3.7.6-23.el9_3.x86_64 17/28
Verifying     : gnutls-3.7.6-23.el9.x86_64 18/28
Verifying     : tzdata-2024a-1.el9.noarch 19/28
Verifying     : tzdata-2023d-1.el9.noarch 20/28
Verifying     : selinux-policy-38.1.23-1.el9_3.2.noarch 21/28
Verifying     : selinux-policy-38.1.23-1.el9_3.1.noarch 22/28
Verifying     : selinux-policy-targeted-38.1.23-1.el9_3.2.noarch 23/28
Verifying     : selinux-policy-targeted-38.1.23-1.el9_3.1.noarch 24/28
Verifying     : sudo-1.9.5p2-10.el9_3.x86_64 25/28
Verifying     : sudo-1.9.5p2-9.el9.x86_64 26/28
Verifying     : rh-amazon-rhui-client-4.0.14-1.el9.noarch 27/28
Verifying     : rh-amazon-rhui-client-4.0.13-1.el9.noarch 28/28
Installed products updated.

Upgraded:
gnutls-3.7.6-23.el9_3.x86_64 nspr-4.35.0-6.el9_3.x86_64 nss-3.90.0-6.el9_3.x86_64 nss-softoken-3.90.0-6.el9_3.x86_64
nss-softoken-freebl-3.90.0-6.el9_3.x86_64 nss-sysinit-3.90.0-6.el9_3.x86_64 nss-util-3.90.0-6.el9_3.x86_64 openssl-1.3.0.7-25.el9_3.x86_64
openssl-lib-1.3.0.7-25.el9_3.x86_64 rh-amazon-rhui-client-4.0.14-1.el9.noarch selinux-policy-38.1.23-1.el9_3.2.noarch selinux-policy-targeted-38.1.23-1.el9_3.2.noarch
sudo-1.9.5p2-10.el9_3.x86_64 tzdata-2024a-1.el9.noarch

Complete!
[ec2-user@ip-172-31-24-127 ~]$ sudo mkdir efs
[ec2-user@ip-172-31-24-127 ~]$ ls
efs
[ec2-user@ip-172-31-24-127 ~]$ sudo yum install nfs-utils -y
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to register.

Red Hat Enterprise Linux 9 for x86_64 - AppStream from RHUI (RPMs) 71 kB/s | 4.5 kB 00:00
Red Hat Enterprise Linux 9 for x86_64 - BaseOS from RHUI (RPMs) 65 kB/s | 4.1 kB 00:00
Red Hat Enterprise Linux 9 Client Configuration 29 kB/s | 1.5 kB 00:00
Dependencies resolved.
=====
Package Architecture Version Repository Size
=====
Installing:
nfs-utils x86_64 1:2.5.4-20.el9 rhel-9-baseos-rhui-rpms 458 k
Installing dependencies:
gssproxy x86_64 0.8.4-6.el9 rhel-9-baseos-rhui-rpms 114 k
keyutils x86_64 1.6.3-1.el9 rhel-9-baseos-rhui-rpms 78 k
=====
```

```
ec2-user@ip-172-31-24-127:~
(10/11): nfs-utils-2.5.4-20.el9.x86_64.rpm 16 MB/s | 458 kB 00:00
(11/11): sssd-nfs-idmap-2.9.1-4.el9_3.5.x86_64.rpm 2.1 MB/s | 44 kB 00:00
-----
Total 7.3 MB/s | 1.2 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : libtirpc-1.3.3-2.el9.x86_64 1/1
  Installing     : libnfsidmap-1:2.5.4-20.el9.x86_64 1/11
  Installing     : rpcbind-1.2.6-5.el9.x86_64 2/11
  Running scriptlet: rpcbind-1.2.6-5.el9.x86_64 3/11
  Installing     : rpcbind-1.2.6-5.el9.x86_64 3/11
  Running scriptlet: rpcbind-1.2.6-5.el9.x86_64 3/11
Created symlink /etc/systemd/system/multi-user.target.wants/rpcbind.service -> /usr/lib/systemd/system/rpcbind.service.
Created symlink /etc/systemd/system/sockets.target.wants/rpcbind.socket -> /usr/lib/systemd/system/rpcbind.socket.

  Installing     : keyutils-1.6.3-1.el9.x86_64 4/11
  Installing     : quota-nls-1:4.06-6.el9.noarch 5/11
  Installing     : quota-1:4.06-6.el9.x86_64 6/11
  Installing     : libev-4.33-5.el9.x86_64 7/11
  Installing     : libverto-libev-0.3.2-3.el9.x86_64 8/11
  Installing     : gssproxy-0.8.4-6.el9.x86_64 9/11
  Running scriptlet: gssproxy-0.8.4-6.el9.x86_64 9/11
  Running scriptlet: nfs-utils-1:2.5.4-20.el9.x86_64 10/11
  Installing     : nfs-utils-1:2.5.4-20.el9.x86_64 10/11
  Running scriptlet: nfs-utils-1:2.5.4-20.el9.x86_64 10/11
  Installing     : sssd-nfs-idmap-2.9.1-4.el9_3.5.x86_64 11/11
  Running scriptlet: sssd-nfs-idmap-2.9.1-4.el9_3.5.x86_64 11/11
  Verifying      : libev-4.33-5.el9.x86_64 1/11
  Verifying      : libverto-libev-0.3.2-3.el9.x86_64 2/11
  Verifying      : quota-1:4.06-6.el9.x86_64 3/11
  Verifying      : quota-nls-1:4.06-6.el9.noarch 4/11
  Verifying      : rpcbind-1.2.6-5.el9.x86_64 5/11
  Verifying      : keyutils-1.6.3-1.el9.x86_64 6/11
  Verifying      : gssproxy-0.8.4-6.el9.x86_64 7/11
  Verifying      : libnfsidmap-1:2.5.4-20.el9.x86_64 8/11
  Verifying      : libtirpc-1.3.3-2.el9.x86_64 9/11
  Verifying      : nfs-utils-1:2.5.4-20.el9.x86_64 10/11
  Verifying      : sssd-nfs-idmap-2.9.1-4.el9_3.5.x86_64 11/11
Installed products updated.

Installed:
gssproxy-0.8.4-6.el9.x86_64      keyutils-1.6.3-1.el9.x86_64      libev-4.33-5.el9.x86_64      libnfsidmap-1:2.5.4-20.el9.x86_64      libtirpc-1.3.3-2.el9.x86_64
libverto-libev-0.3.2-3.el9.x86_64  nfs-utils-1:2.5.4-20.el9.x86_64  quota-1:4.06-6.el9.x86_64  quota-nls-1:4.06-6.el9.noarch      rpcbind-1.2.6-5.el9.x86_64
sssd-nfs-idmap-2.9.1-4.el9_3.5.x86_64

Complete!
[ec2-user@ip-172-31-24-127 ~]$
```

```
Installed:
gssproxy-0.8.4-6.el9.x86_64      keyutils-1.6.3-1.el9.x86_64      libev-4.33-5.el9.x86_64      libnfsidmap-1:2.5.4-20.el9.x86_64      libtirpc-1.3.3-2.el9.x86_64
libverto-libev-0.3.2-3.el9.x86_64  nfs-utils-1:2.5.4-20.el9.x86_64  quota-1:4.06-6.el9.x86_64  quota-nls-1:4.06-6.el9.noarch      rpcbind-1.2.6-5.el9.x86_64
sssd-nfs-idmap-2.9.1-4.el9_3.5.x86_64

Complete!
[ec2-user@ip-172-31-24-127 ~]$ sudo mount -t nfs4 -o nfsvers=4.1,rsz=1048576,wsz=1048576,hard,timeo=600,retrans=2,noresvport fs-021c5a2d2f5710cc4.efs.us-east-1.amazonaws.com:/ efs
[ec2-user@ip-172-31-24-127 ~]$ df -h
Filesystem              Size  Used Avail Use% Mounted on
devtmpfs                4.0M   0  4.0M   0% /dev
tmpfs                   377M   0  377M   0% /dev/shm
tmpfs                   151M  4.3M  147M   3% /run
/dev/xvda4              9.2G  1.5G   7.8G  16% /
/dev/xvda3              536M 161M  376M  31% /boot
/dev/xvda2              200M  7.0M  193M   4% /boot/efi
tmpfs                   76M   0   76M   0% /run/user/1000
fs-021c5a2d2f5710cc4.efs.us-east-1.amazonaws.com:/ 8.0E   0  8.0E   0% /home/ec2-user/efs
[ec2-user@ip-172-31-24-127 ~]$
```